Predicting Tropospheric Ozone Levels from Weather and Precursor Pollutants

UE24CS352A Machine Learning, Mini-Project · Replication of Sutton, CS229 Autumn 2019

1. Problem Statement

Ground-level ozone is a harmful pollutant formed when NO_x and volatile organic compounds (VOCs) react in sunlight. The US EPA designates regions that exceed the ozone standard as "nonattainment", which triggers costly regulation. Forecasting ozone lets a region act early. The task is a regression problem: given a day's **temperature, pressure, wind speed, month, location, NO_2 and VOC levels** at a monitoring site, predict that day's ozone concentration (ppm). We replicate the original paper, which compared linear regression, support vector regression (SVR) and a neural network, and then try simple improvements.

2. Dataset

The paper used the *EPA Historical Air Quality* dataset on Google BigQuery. We use the same source published as EPA AirData daily summary files (no cloud account needed), 1980–2017, about 600 MB across six files per year: ozone (label), NO_2, VOCs, temperature, pressure and wind.

- Each file is averaged per *(site, date)*. For VOCs we average over all measured species, and for wind we keep only wind speed. The six tables are inner-joined so each row is one site on one day with every feature present. Latitude/longitude are used for location instead of the numeric site code.
- Final dataset: **90,997 rows** from sites that measure all six quantities. Split by time as in the paper: **train 1980–2016 (86,461)**, **validation Jan–Jun 2017 (2,384)**, **test Jul–Dec 2017 (2,152)**.
- The paper reports about 2× more rows (200,652 train). We believe this is because the BigQuery tables contain several rows per site-day (multiple monitors / ozone standards), which we merge into one.
- Features are min-max scaled to [0,1] using statistics from the training set only.

3. Approach

- **Linear Regression** (ordinary least squares), as the baseline.
- **SVR, paper setting:** polynomial kernel with sklearn's default ε = 0.1. Kernel SVR scales roughly quadratically with data size, so all SVRs are trained on a random 20,000-row subset.
- **SVR, fixed:** the paper's own post-mortem says ε was too large. Ozone values are around 0.03 ppm, so a 0.1 ppm tube contains every point and the model learns nothing. We use ε = 0.002 with an RBF kernel.
- **Neural Network:** MLP with 3 hidden layers of 200 ReLU units and an adaptive learning rate, as in the paper, trained with Adam and early stopping.
- **Gradient Boosting** (our addition): sklearn's `HistGradientBoostingRegressor`, which is fast, needs no scaling and works well on tabular data.
- Models are compared with R^2 (as in the paper) and mean absolute error. Permutation feature importance is used to answer the paper's "future work" question of which features matter most.

4. Implementation

Python with pandas, scikit-learn and matplotlib, organised as a small pipeline: `download.py` (parallel download of EPA zips) → `build_dataset.py` (cleaning, per-site-day aggregation, joins) → `train.py` (time-based split, scaling, training, metrics, saving models) → `evaluate.py` (plots, feature importance) → `predict.py` (single prediction plus the EPA AQI category, used for the live demo). The full pipeline runs in under 10 minutes on a laptop once the data is downloaded.

5. Results

Model	Paper R^2		Our R^2			Test MAE (ppm)
	Train	Test	Train	Val	Test	
Linear Regression	0.489	0.467	0.339	0.204	0.412	0.0073

SVR (poly, ε=0.1)	−0.319	−2.363	−1.702	−2.115	−2.508	0.0198
SVR (RBF, ε=0.002)	–	–	0.607	0.563	0.612	0.0059
Neural Network	0.702	0.654	0.601	0.519	0.555	0.0064
Gradient Boosting	–	–	0.708	0.577	**0.649**	**0.0056**

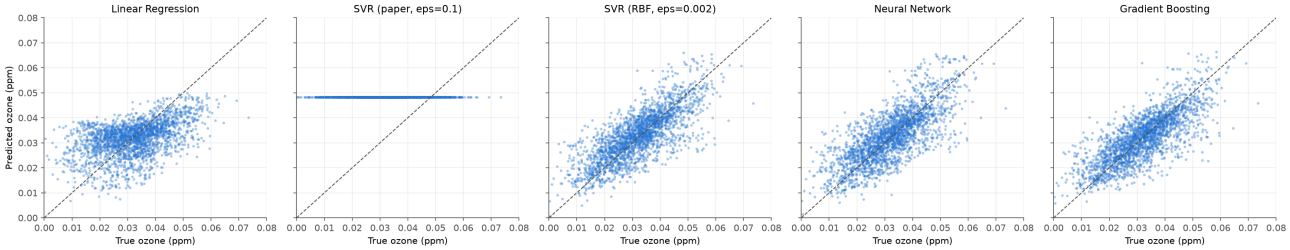


Figure 1. True vs predicted ozone on the validation set (Jan–Jun 2017). The dashed line is a perfect prediction.

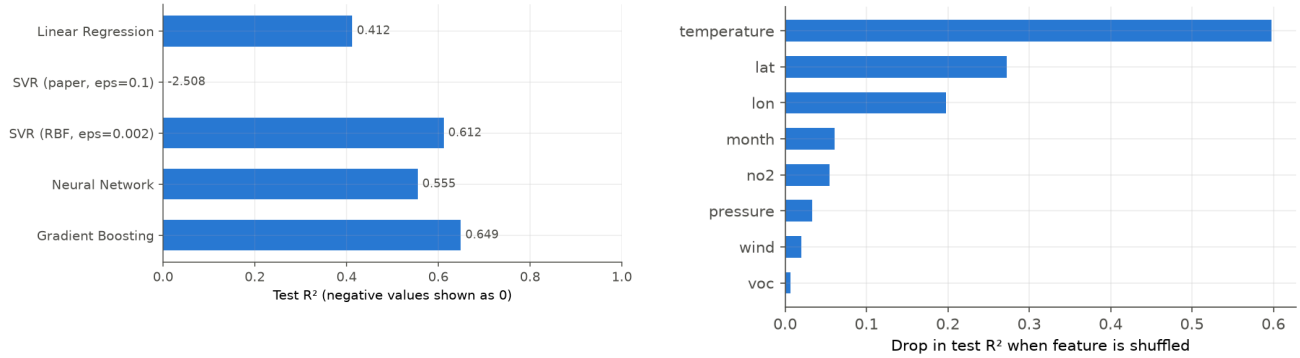


Figure 2. Left: test R^2 per model. Right: permutation importance for gradient boosting on the test set.

The paper's ranking replicates. Linear regression is weak ($R^2 \approx 0.4$) because ozone depends non-linearly on its inputs. The paper-setting SVR outputs an almost constant value (the flat line in Figure 1) and scores a strongly negative R^2, just like the paper's −2.36. Lowering ε alone turns it into one of the best models (0.61). Our neural network scores 0.555, below the paper's 0.65, likely because our de-duplicated dataset is half the size and early stopping ends training after 20 epochs. Gradient boosting gives the best test R^2 (0.649) and MAE, needs no feature scaling, and trains in seconds rather than minutes. Temperature is by far the most important feature (a proxy for sunlight), followed by location; the precursors NO_2 and VOC matter less than expected once weather and location are known.

6. Conclusions

- We reproduced the paper's main findings using the same public EPA data: linear models underfit, an SVR with too large an ε fails completely, and non-linear models reach $R^2 \approx 0.6$–0.65.
- Two simple changes beat or match the paper's best model: tuning SVR's ε to the scale of the target, and using gradient-boosted trees, which are cheaper and easier to train than a deep MLP.
- Validation R^2 is lower than test R^2 for every model, because Jan–Jun has a smaller spread of ozone values. Plain R^2 on a single six-month window is therefore a noisy measure.
- Limitations / future work: only 86 sites (36 in 2017) measure every feature; adding sources such as nearby traffic or industry (e.g. Census data), using the previous day's ozone, and predicting the EPA nonattainment class directly as a classification problem.